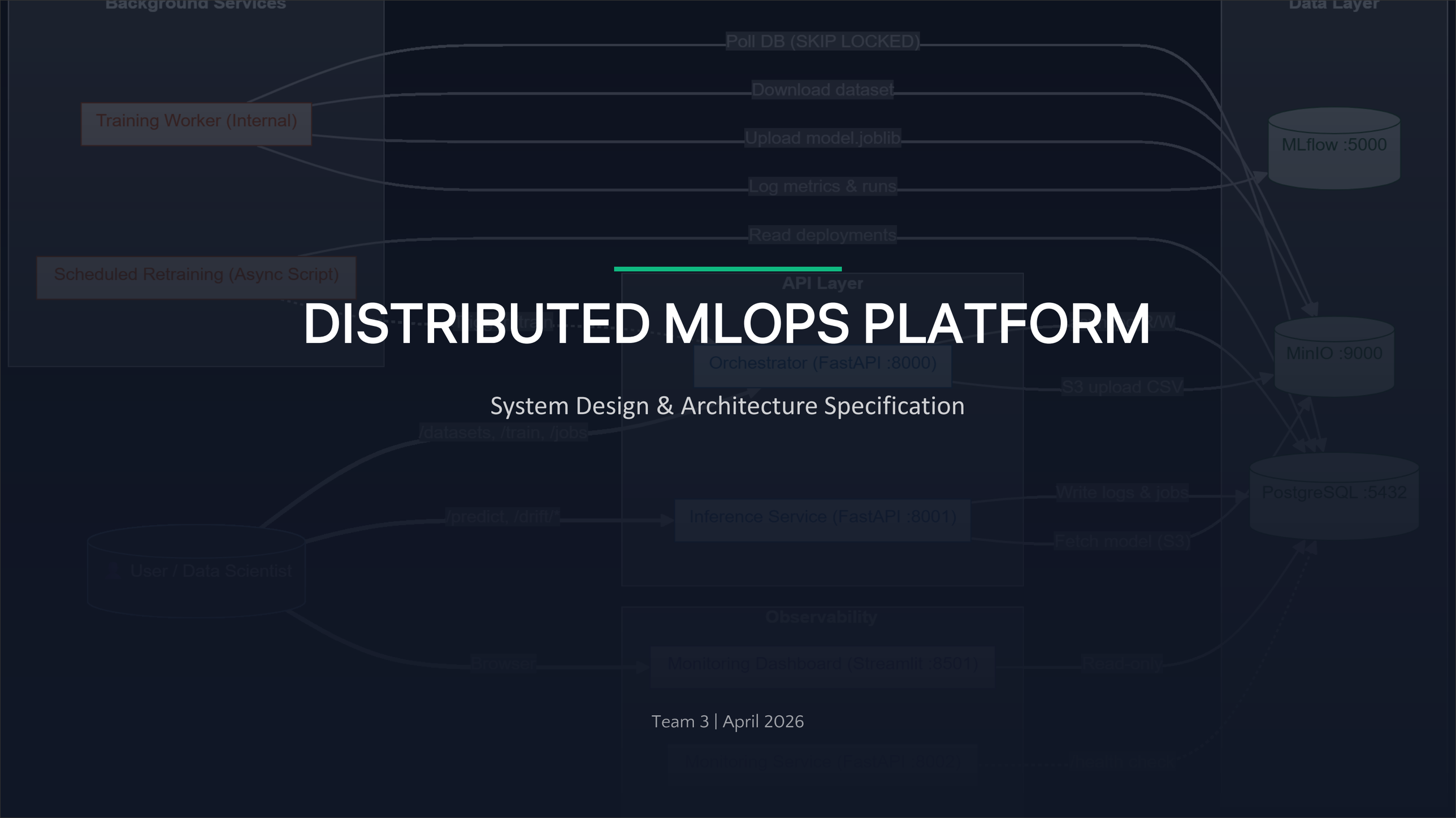


DISTRIBUTED MLOPS PLATFORM

System Design & Architecture Specification

Team 3 | April 2026

What We'll Cover

01

The Problem

Why ML models get stuck in notebooks and never reach production

02

System Architecture

How the platform is built and how components interact

03

Key Features

Training, inference, storage, and monitoring capabilities

04

Roadmap & Team

Where we're headed and who built this platform

01

The Problem

Why we built this platform and what it solves

From Notebook to Production

The Challenge

ML models often get stuck in experimental environments and never reach production.

The Pain Points

- Fractured data pipelines
- Lost model traceability
- Scalability limits (Python GIL)

Our Solution

A lightweight infrastructure managing the full ML lifecycle — from automated training to optimized inference.

Before vs. After

BEFORE

Manual model tracking in notebooks

No versioning or reproducibility

Hand-off gaps between data scientists and engineers

Deployments are fragile and error-prone

AFTER

Automated training pipeline with state tracking

Full model lineage via MLflow + MinIO

One platform for scientists and engineers

Resilient, observable, production-ready

02

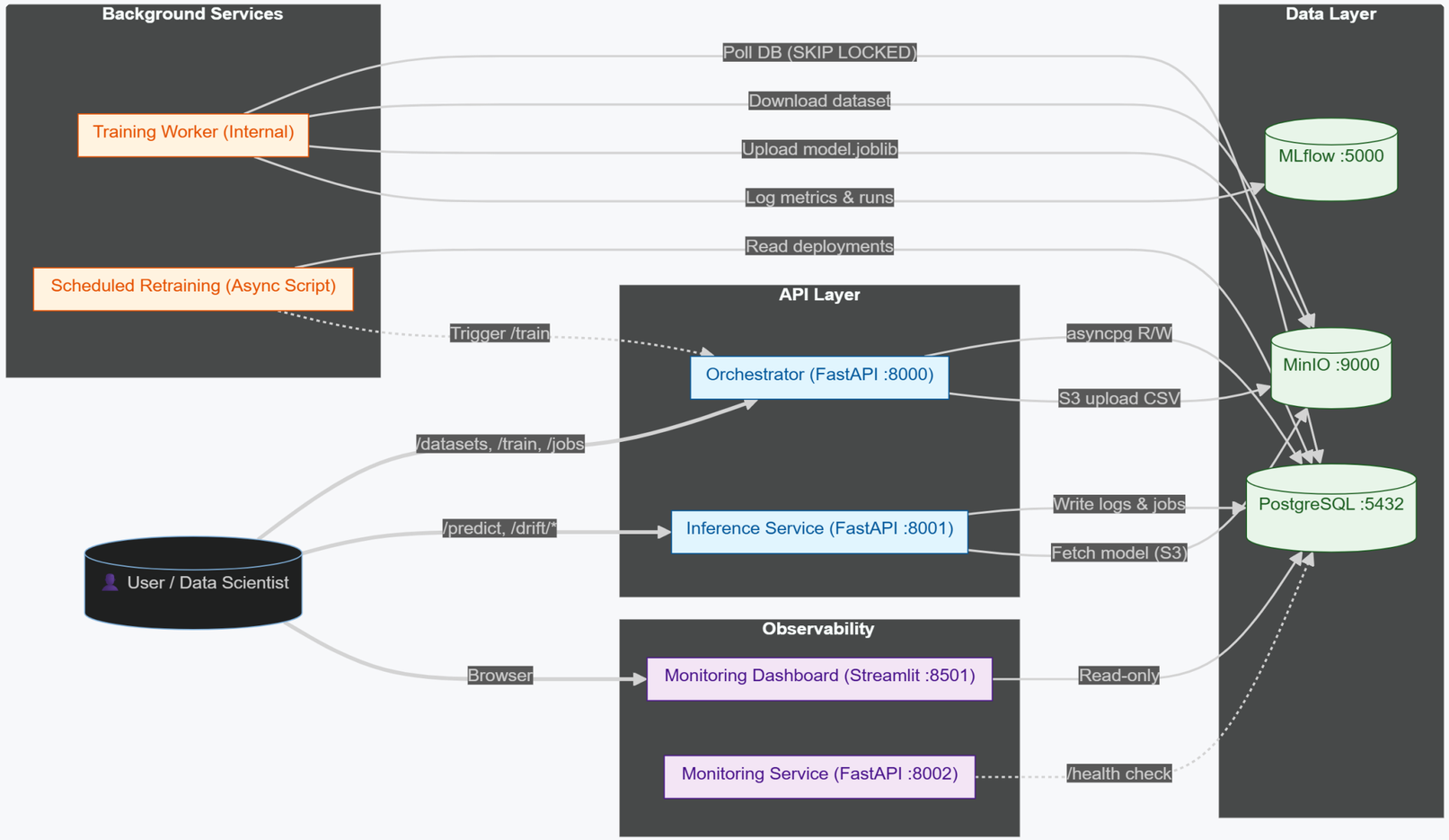
System Architecture

How the platform is built and how components interact

interact

Three-Plane Architecture

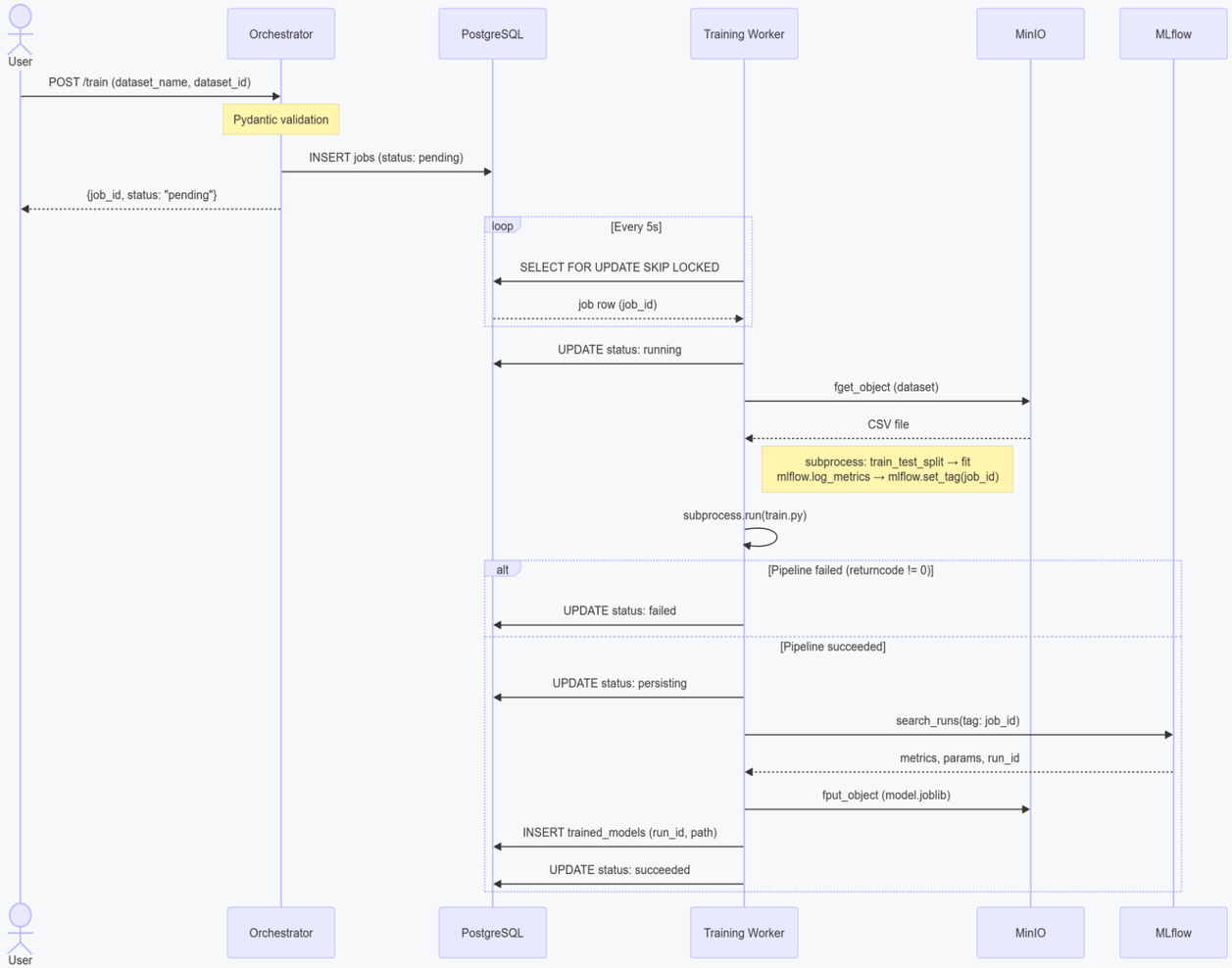
9 containerized services running on Docker Compose



Control Plane | Execution Plane | Storage Plane

Model Lifecycle

From raw data to live inference in 7 steps



1. Upload dataset via API
2. Request training job
3. Worker polls & atomically locks job
4. Subprocess executes ML pipeline
5. Metrics logged to MLflow
6. Model persisted to MinIO
7. Job finalized as succeeded

Resilience: If a worker crashes, stuck jobs auto-recover to pending state on next startup.

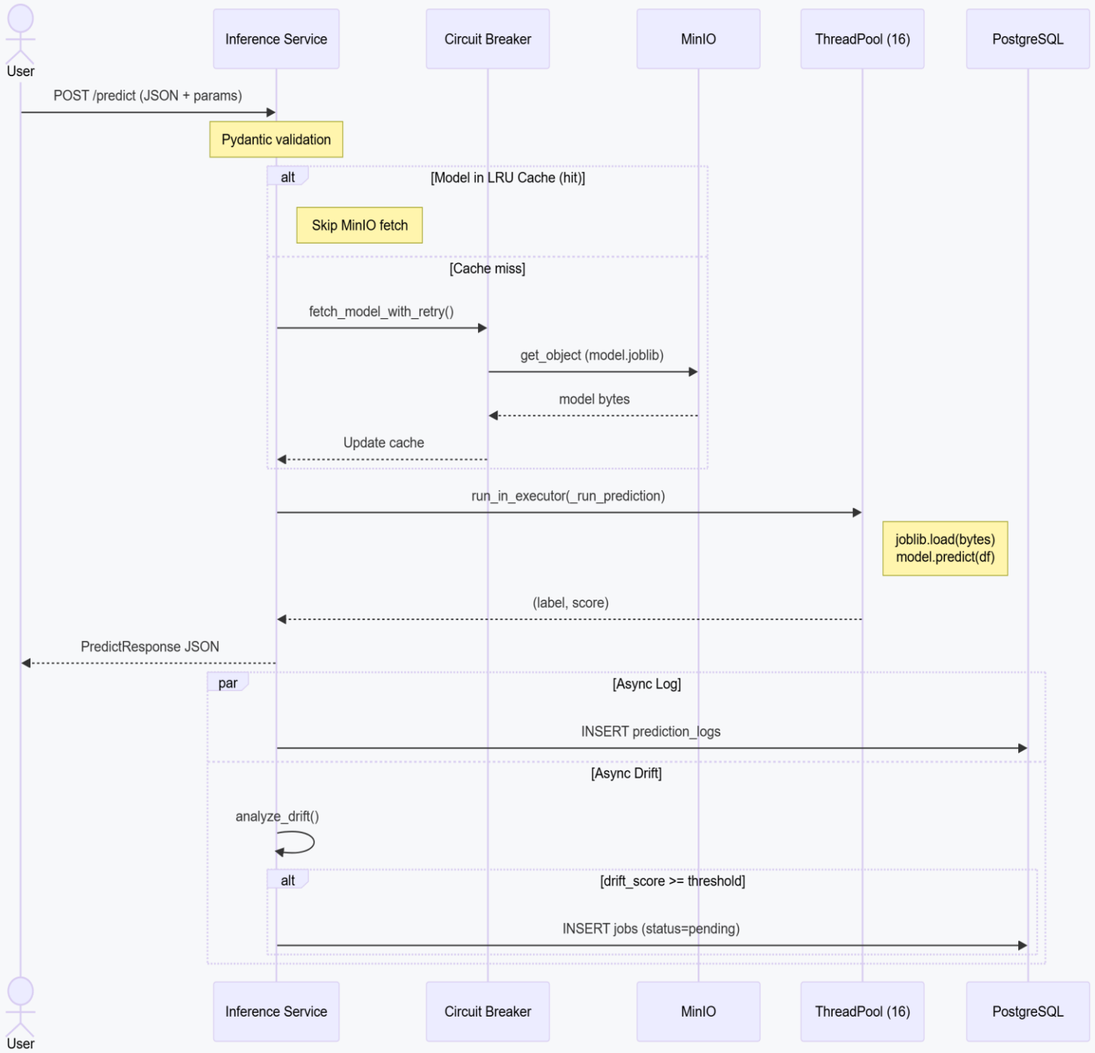
03

Key Features

Training, inference, storage, and monitoring capabilities

capabilities

Smart Training & Fast Inference



Training Worker

OS-level subprocess isolation prevents memory crashes

Scikit-Learn pipeline with MLflow experiment tracking

Auto-recovery for stuck jobs on worker restart

Inference Service

16-thread ThreadPool for concurrent predictions

In-process model caching + per-thread deserialization

Circuit Breaker for MinIO resilience

Storage & Monitoring

MinIO Storage

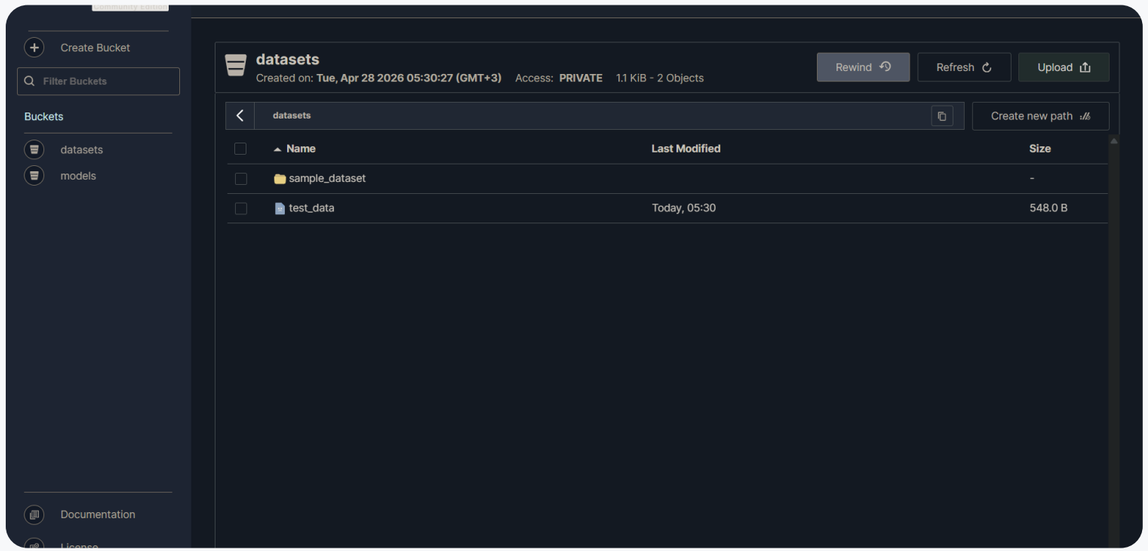
S3-compatible object storage with dedicated buckets for datasets and serialized models

MLflow Tracking

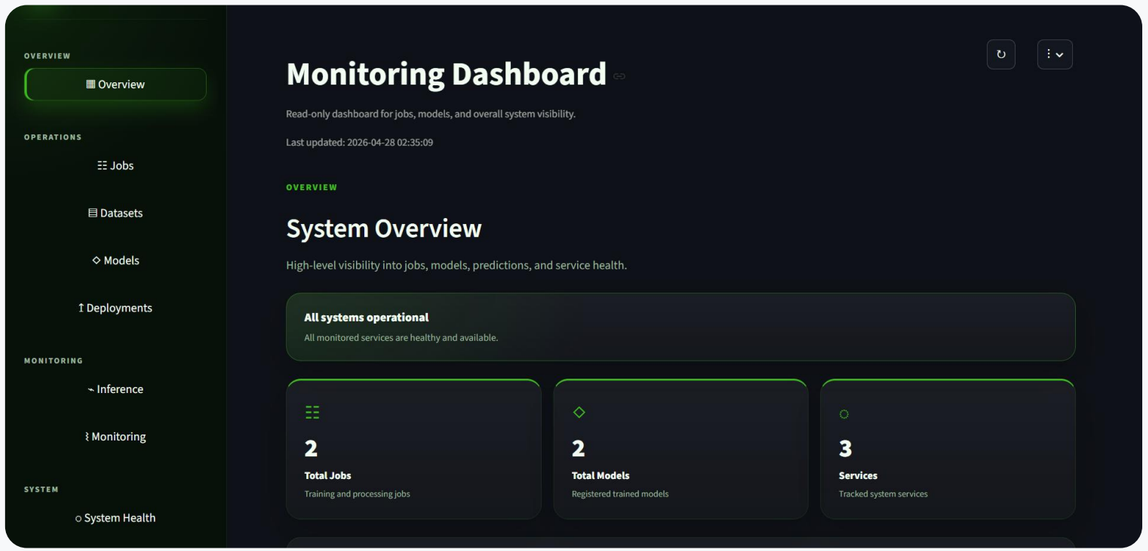
Experiment registry logging Accuracy, Precision, Recall, F1 — all linked to job IDs

Streamlit Dashboard

8-page monitoring app: Overview, Jobs, Datasets, Models, Deployments, Inference, Health



MinIO Object Store



Dashboard System Overview

04

Roadmap & Team

Where we're headed and who built it

Meet the Team

8 engineers, 3 weeks, one platform

Andrei P.

Team Lead
CI/CD, Architecture

Arina S.

Project Manager
Coordination

Elizaveta K.

Orchestrator
Database Design

Ahmed L.

Dashboard

Aleksei

Inference Service
Circuit Breaker

Danil K.

Training Worker
MLflow Integration

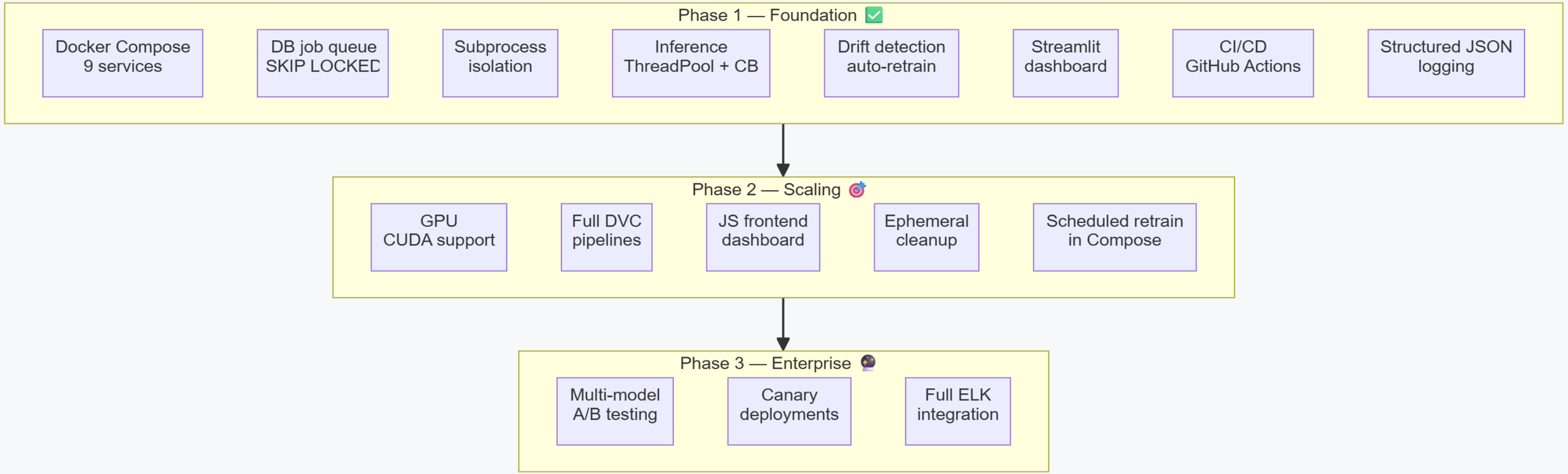
Daria G.

Training Pipelines
Dataset Registration

Julia K.

Reports
Presentations

Development Roadmap



Phase 1 — Foundation

Docker Compose, async jobs, inference, dashboard, CI/CD — complete

Phase 2 — Scaling

GPU support, DVC data lineage, JS frontend, ephemeral cleanup

Phase 3 — Enterprise

Multi-model A/B testing, canary deployments, full ELK integration

THANK YOU

Questions?

Team 3 | Distributed MLOps Platform